



PRAYAS

2.0 2027

Chemistry

Lecture – 1

Redox Reactions

By – Rahul Dudi Sir





Topics to be covered



- 1 Questions based on Mixtures
- 2 Contraction in volume
- 3 Eudiometry ✓
- 4 Redox reactions



Join My Telegram Channel



R

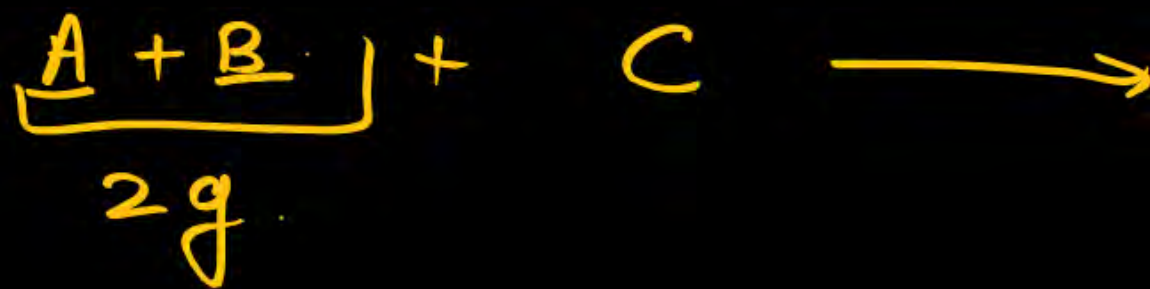


@RAHULDUDICHEMISTRY





Problems Related with Mixture



QUESTION

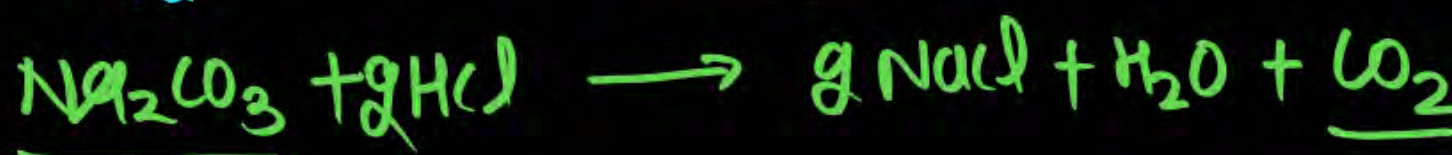
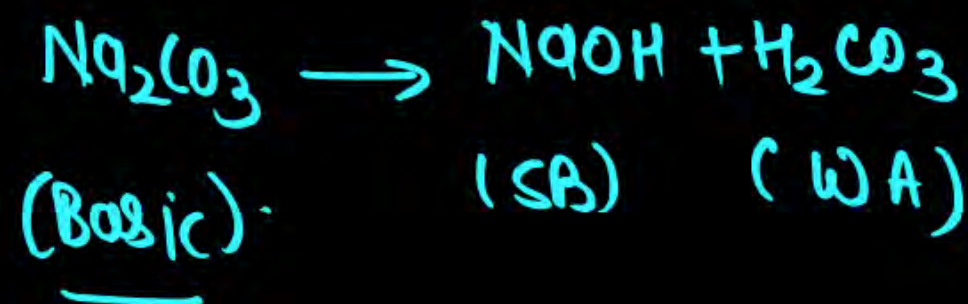
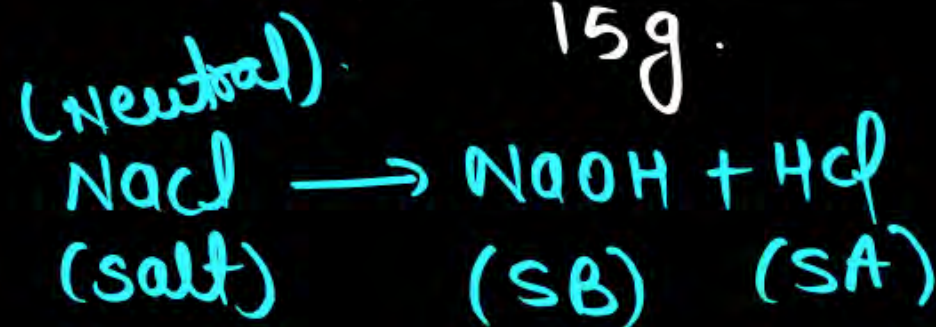
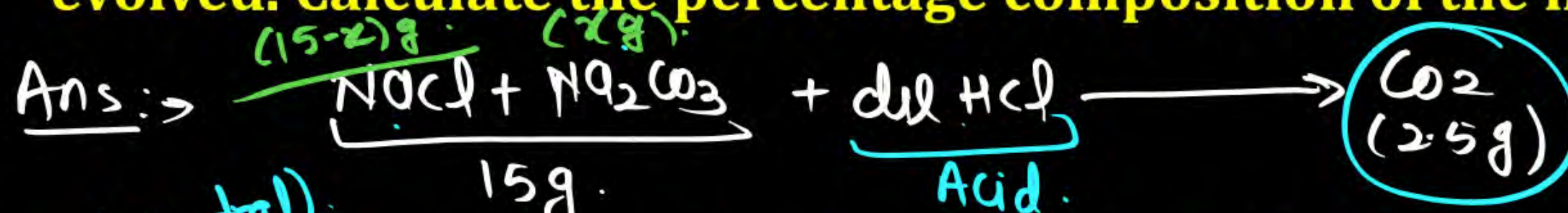
①

Done

MW = 44 g/mol

PW

15 g mixture of NaCl and Na_2CO_3 is heated with dilute HCl, 2.5 g of CO_2 is evolved. Calculate the percentage composition of the mixture.



$$n_{\text{Na}_2\text{CO}_3} = n_{\text{CO}_2}$$

$$\frac{x}{106} = \frac{2.5}{44}$$

$$x = \underline{\underline{6g}}$$

$$\% \text{Na}_2\text{CO}_3 = \frac{6}{15} \times 100\% \approx 40\%$$

$$\% \text{NaCl} \approx 60\%$$

QUESTION

②

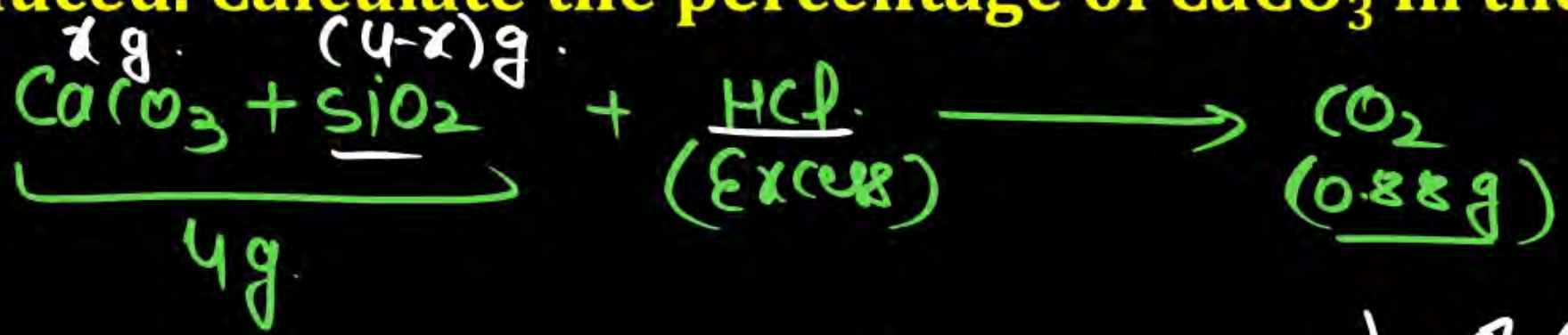
$\text{SiO}_2 \rightarrow \text{Acidic}$ $M_w = 100 \text{ g/mol}$

Done ✓



4 g of a mixture of CaCO_3 and SiO_2 is treated with an excess of HCl and 0.88 g of CO_2 is produced. Calculate the percentage of CaCO_3 in the mixture.

Ans:→



$$n_{\text{CaCO}_3} = n_{\text{CO}_2}$$

$$\frac{x}{100} = \frac{0.88}{44}$$

$$x = 2 \text{ g}$$

$$\begin{aligned} \% \text{CaCO}_3 &= \frac{2}{4} \times 100 \% \\ &= 50 \% \text{ Ans.} \end{aligned}$$

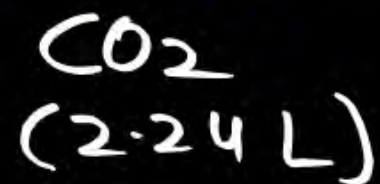
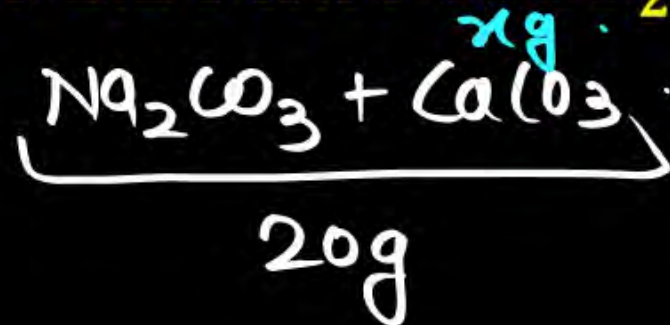
QUESTION

3



20 g mixture of Na_2CO_3 and CaCO_3 is gently heated. On heating this sample produces 2.24 litre of CO_2 at STP. Find the % weight of Na_2CO_3 in that sample.

Ans:-



$$n_{\text{CaCO}_3} = n_{\text{CO}_2}$$

$$\frac{x}{100} = \frac{2.24}{22.4}$$

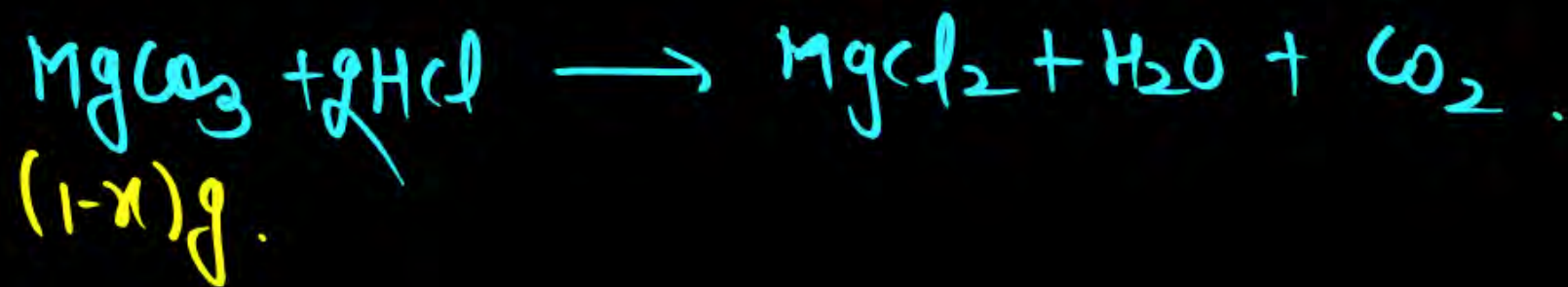
$$\underline{x = 10\text{g}}$$

$$\% \text{CaCO}_3 = \frac{10}{20} \times 100\%$$

$$= \underline{50\%}$$

QUESTION4NW

1 gm of a mixture of CaCO_3 and MgCO_3 is reacted with excess dilute HCl . Thus produced CO_2 whose volume is 240 ml at STP. Calculate the % composition of the mixture. ($P=1\text{ atm}$)





Ans. [CaCO_3 – .625 g, MgCO_3 – 0.375 g]

QUESTION

5 HW



0.931 g of a mixture of NaCl and KCl on treatment with H_2SO_4 gave 1.0784 g of a mixture of Na_2SO_4 and K_2SO_4 . Calculate % composition of the original mixture.



Ans. [NaCl – 57.3%, KCl – 42.64%]



Contraction in volume:

→ Combustion of Hydrocarbon.



Done



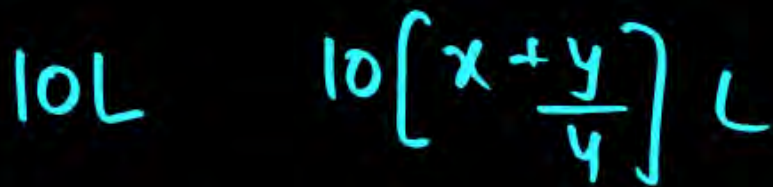
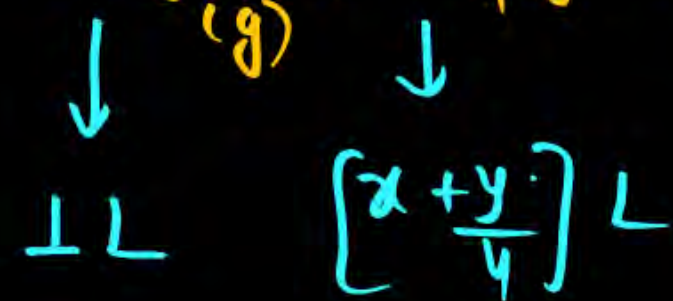
Contraction in volume = Vol. of Reactants - Vol. of Products.

↓
Vol. of C_xH_y + Vol. of O_2

↓
Vol. of CO_2 + Vol. of H_2O

Ex: →

10L



$$\text{Contraction in volume} = \left\{ 10 + 10\left[x + \frac{y}{4}\right] \right\} - 10x$$

$$= \left[10 + 10\frac{y}{4} \right] L \quad \text{Ans}$$



QUESTION

6



Find the volume contraction if 10L of Butane (C_4H_{10}) under combustion with sufficient amount of oxygen?

1 25 ☐

2 35 ☒

3 45 ☐

4 55 ☐



1 L

$\frac{13}{2} L$

4 L

10 L

$\frac{13}{2} \times 10^5 L = 65 L$

$10 \times 4 = 40 L$

Contraction in vol. $= (10 + 65) - 40$
 $= \underline{\underline{35 L}}$

QUESTION

7 HW



1 litre of gaseous aliphatic compound $C_xH_yO_z$ is completely burnt in excess of O_2 and cooled to room temperature. The contraction in volume is:



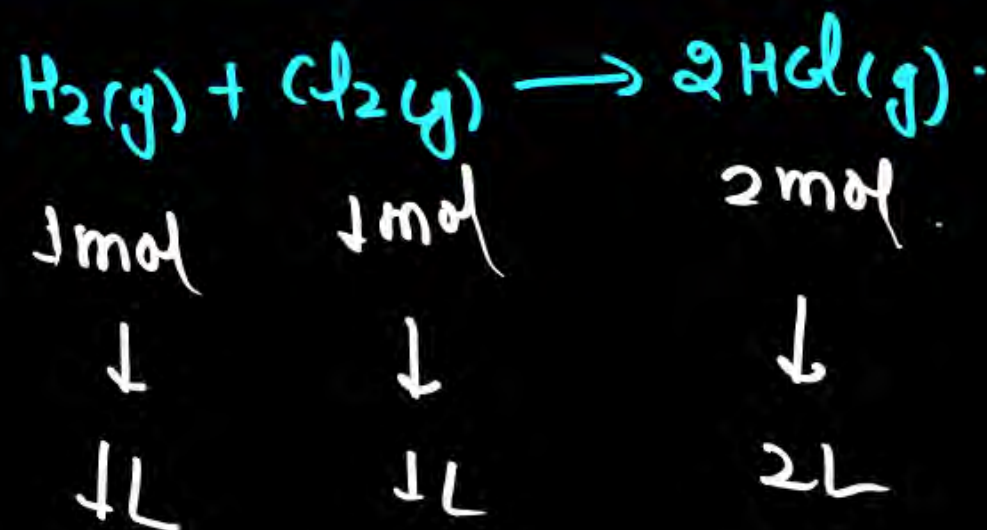
Eudiometry



→ Volumetric Analysis → vol. of gases

Eudiometry or "gas analysis" involves calculations based on gaseous reactions in which at least two components are gases and their amount is given in terms of volumes measured at same pressure & Temperature.

* Gay-Lussac's law of gaseous vol.





What can we find using Eudiometry



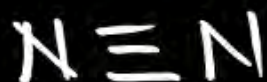
- **mol. formula of Hydrocarbon** ✖
- **composition of gaseous mixture** ✖
- **vol. -vol. relationship amongst gases or simple gaseous reactions.**
- **mol. formula of gases.**

* Combustion of Hydrocarbon:



* Contraction in Volume

* $\text{N}_2(g)$ is considered Inert



* $\text{N}_2 + \text{O}_2 \not\rightarrow \text{oxides of nitrogen.}$





Eudiometry

याद रखनी है



O_3	Absorbed by	Turpentine oil
O_2	Absorbed by	Alkaline pyrogallol
NO	Absorbed by	$FeSO_4$ Solution
Cl_2, CO_2, SO_2	Absorbed by	NaOH, KOH, $Ca(OH)_2$
NH_3, HCl	Absorbed by	Water (l)
H_2O (vapours)	Absorbed by	<u>Anhy. $CuSO_4$, Anhy. $CaCl_2$, Conc H_2SO_4</u>
CO	Absorbed by	<u>Ammoniacal Cu_2Cl_2</u>

QUESTION

Notes



Find volume of C_3H_8 Which will react with 30 L of O_2 .

1

3

2

6

3

9

4

None of these

20 mL of a hydrocarbon required 100 mL of O_2 for complete combustion and 60 mL of CO_2 is formed. The formula of the hydrocarbon is:

1 C_3H_6

2 C_2H_4

3 C_3H_8

4 C_2H_6



$$1 \text{ mL} \quad \left(x + \frac{y}{4} \right) \text{ mL} \quad x \text{ mL}$$

$$20 \text{ mL} \quad 20 \left[x + \frac{y}{4} \right] \text{ mL} \quad \downarrow \quad 20x \text{ mL}$$

$$20x = 60$$

$$x = 3$$



Ans. [3]

QUESTION

Notes



10 mL of gaseous hydrocarbon on complete combustion gives 40 mL of CO_2 and 50 mL of H_2O (Vapour). The hydrocarbon is:

1 C_3H_6

2 C_2H_4

3 C_3H_8

4 C_4H_{10}

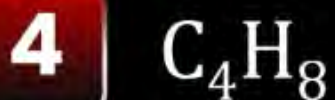
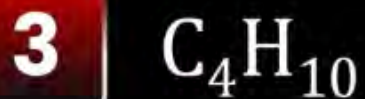
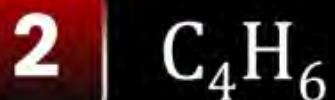


Ans. [4]



At 300 K and 1 atmospheric pressure, 10 mL of a hydrocarbon required 55 mL of O_2 for complete combustion and 40 mL of CO_2 is formed. The formula of the hydrocarbon is:

[2019 Main, 10 April-I]



QUESTIONNotes

15 ml of a gaseous hydrocarbon required 357.15 ml of air for complete combustion. The air contains 21% by volume of oxygen. The volume of gaseous mixture was found to be 327 ml (all volumes being measured at STP). what is the formula of the hydrocarbon:

1**2****3****4**

QUESTION

Notes



3 litre of mixture of propane and butane on complete combustion give 10 litre CO_2 . find the composition of the mixture.

20 ml of a gaseous hydrocarbon required 400 ml of air for complete combustion. The air contains 20% by volume of oxygen. The volume of gaseous mixture after explosion and cooling was found to be 380 ml. volume of CO_2 is:

1 40 ml

2 60 ml

3 80 ml

4 100 ml



Ans. [2]



Question

8 HW



A gaseous hydrocarbon requires 6 times its own volume of O_2 for complete oxidation and produces 4 times its volume of CO_2 . What is its formula?



Ans. [C_4H_8]

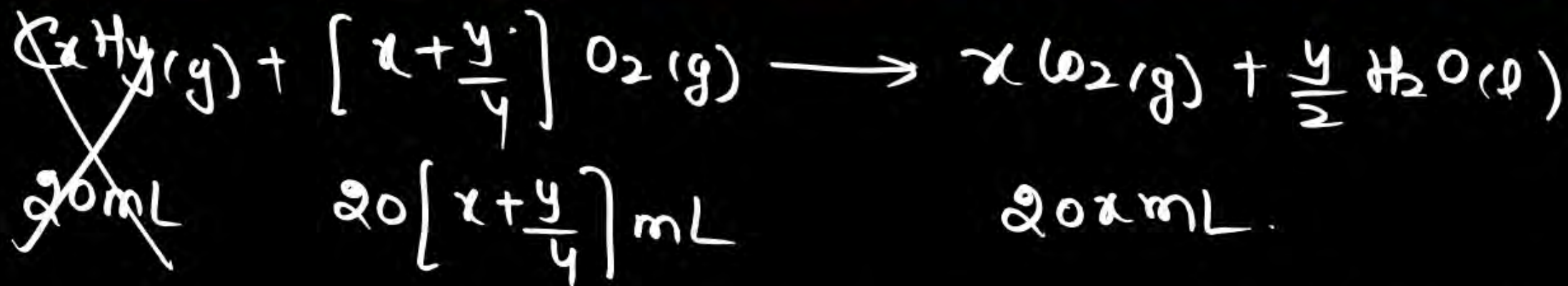
Question

9



A 20 ml sample of hydrocarbon on treatment with excess of $O_2(g)$ observed a volume contraction of 60 ml. on passing through KOH solution, another contraction of 80ml is observed. find molecular formula of the hydrocarbon.

Ans:→



$$V_{CO_2} = 80$$

$$20x = 80$$

$$\boxed{x=4}$$

$$Vol. contraction = V_{Reactants} - V_{Products}$$

$$20 + 20 \left[x + \frac{y}{4} \right] - V_{CO_2} = 60$$

$$20 \left[x + \frac{y}{4} \right] = 120$$

$$x + \frac{y}{4} = 6$$

$$\boxed{y=8}$$

$$formula = C_4H_8$$

Ans.

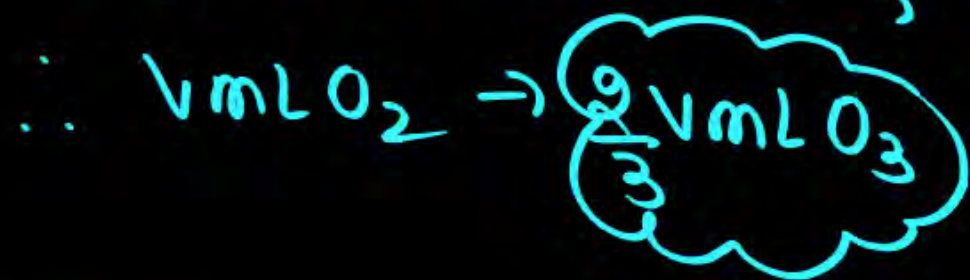
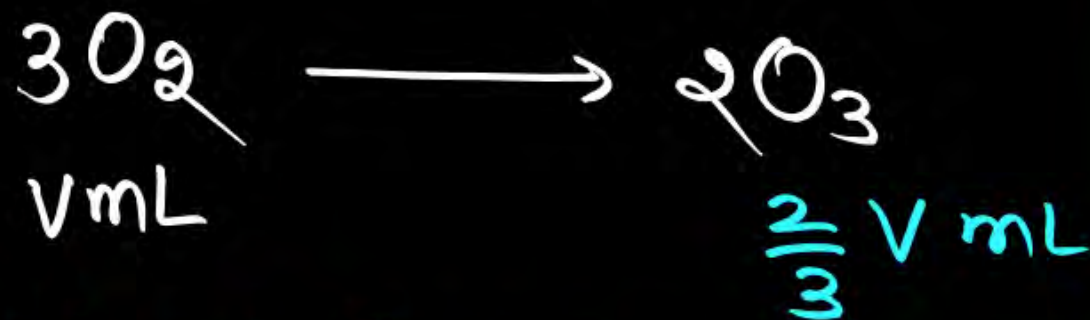
Question

10
1000 mL → Total vol.



1 litre of O_2 gas is passed through an ozonizer and the final volume of gas mixture is found to be 820 ml. If this gas mixture is passed through turpentine oil, find volume of gas remaining.

Ans: → Ozonizer →



$$(V_{O_2})_{\text{left}} = (1000 - V) \text{ ml}$$

$$(V_{O_2})_{\text{left}} + (V_{O_3})_{\text{formed}} = 820 \text{ ml}$$

$$1000 - V + \frac{2}{3}V = 820$$

$$\boxed{V = 540 \text{ mL}}$$

$$(V_{O_2})_{\text{left}} = 1000 - V$$

$$= 1000 - 540$$

$$\Rightarrow \underline{\underline{460 \text{ mL}}}$$

Question

11

HW



A sample of water gas has composition by volume of 50% H_2 , 45% CO and 5% CO_2 gases. find volume in litre of water gas which on treatment with excess of steam will produce 10 litre H_2 under same T and P.

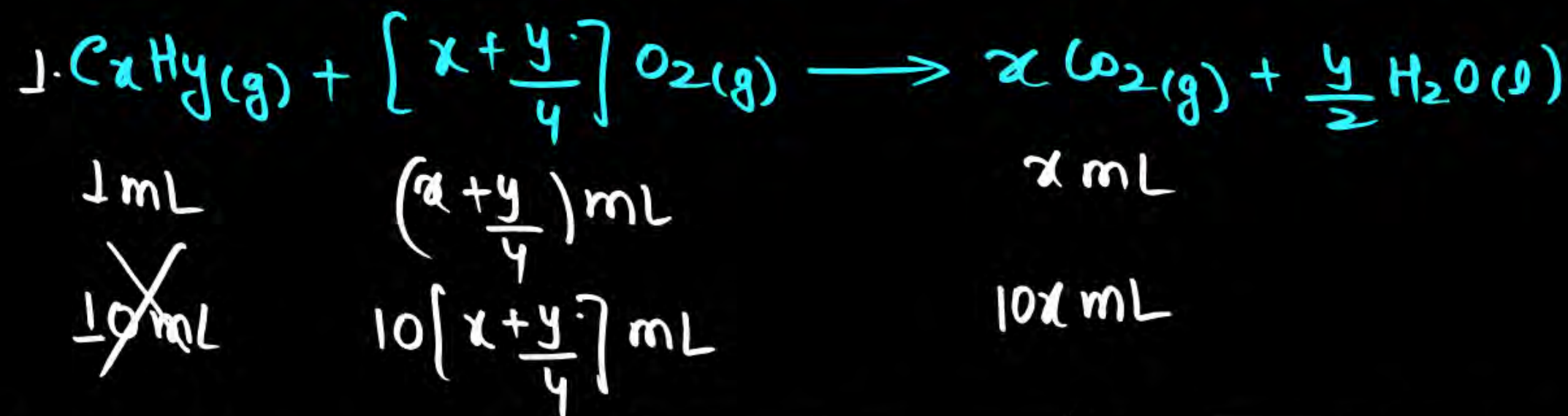
Question

12



10 ml of a gaseous hydrocarbon were burnt completely in 80 ml of O_2 at STP. The remaining gas occupied 70 ml at STP. This volume becomes 50 ml on treatment with KOH solution. What is the empirical formula of hydrocarbon?

Ans:→



* $(V_{O_2})_{left} + V_{CO_2} = 70 \text{ ml}$

$V_{CO_2} = 20 \text{ ml}$

$(V_{O_2})_{left} = \underline{50 \text{ ml}}$

$10x = 20$

$\underline{x = 2}$

$(V_{O_2})_{reacted} = 80 - 50 = 30$

$10 \left[x + \frac{y}{4} \right] = 30$

$\underline{y = 4}$

formula

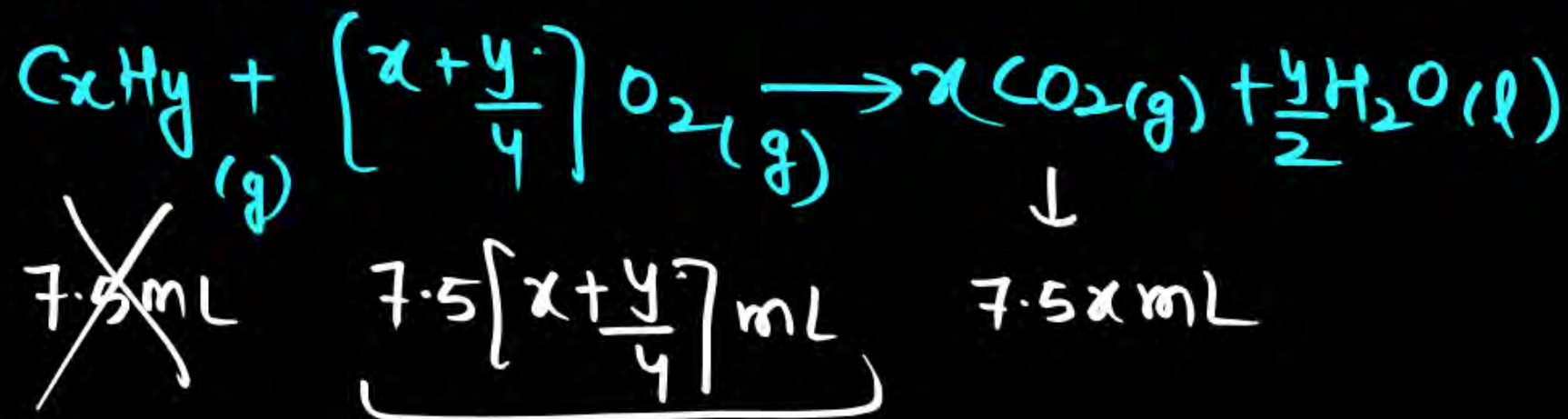


Ethene

Ans

7.5 ml of a gaseous hydrocarbon was exploded with 36 ml of O_2 . On cooling the volume of gases was found to be 28.5 ml, 15 ml of which was absorbed by KOH and the rest was absorbed in a solution of alkaline pyrogallol. Deduce formula of hydrocarbon.

Ans:→



$$7.5 \left[x + \frac{y}{4}\right] = 22.5$$

$$y = 4$$

formula → C_2H_4 (Ans)

$O_2(g)$

$$V_{CO_2} = 15 \text{ ml}$$

$$7.5x = 15$$

$$x = 2$$

$$(V_{O_2})_{\text{left}} = 28.5 - 15 = 13.5 \text{ ml}$$

$$(V_{O_2})_{\text{reacted}} = (36 - 13.5) \text{ ml} = 22.5 \text{ ml}$$

QUESTION

14
HW



80 mL of a hydrocarbon on mixing with 264 mL of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL. When the system is treated with KOH solution, the volume decreases to 64 mL. The formula of the hydrocarbon is:

[JEE Main 2026 (21 January Shift-I)]

1 C_2H_2

2 C_2H_4

3 C_2H_6

4 C_4H_{10}



Ans: [1]



Ar samajh aa raha
hai ya
nahi??

Physics Wallah

Redox Reactions

Reduction + Oxidation

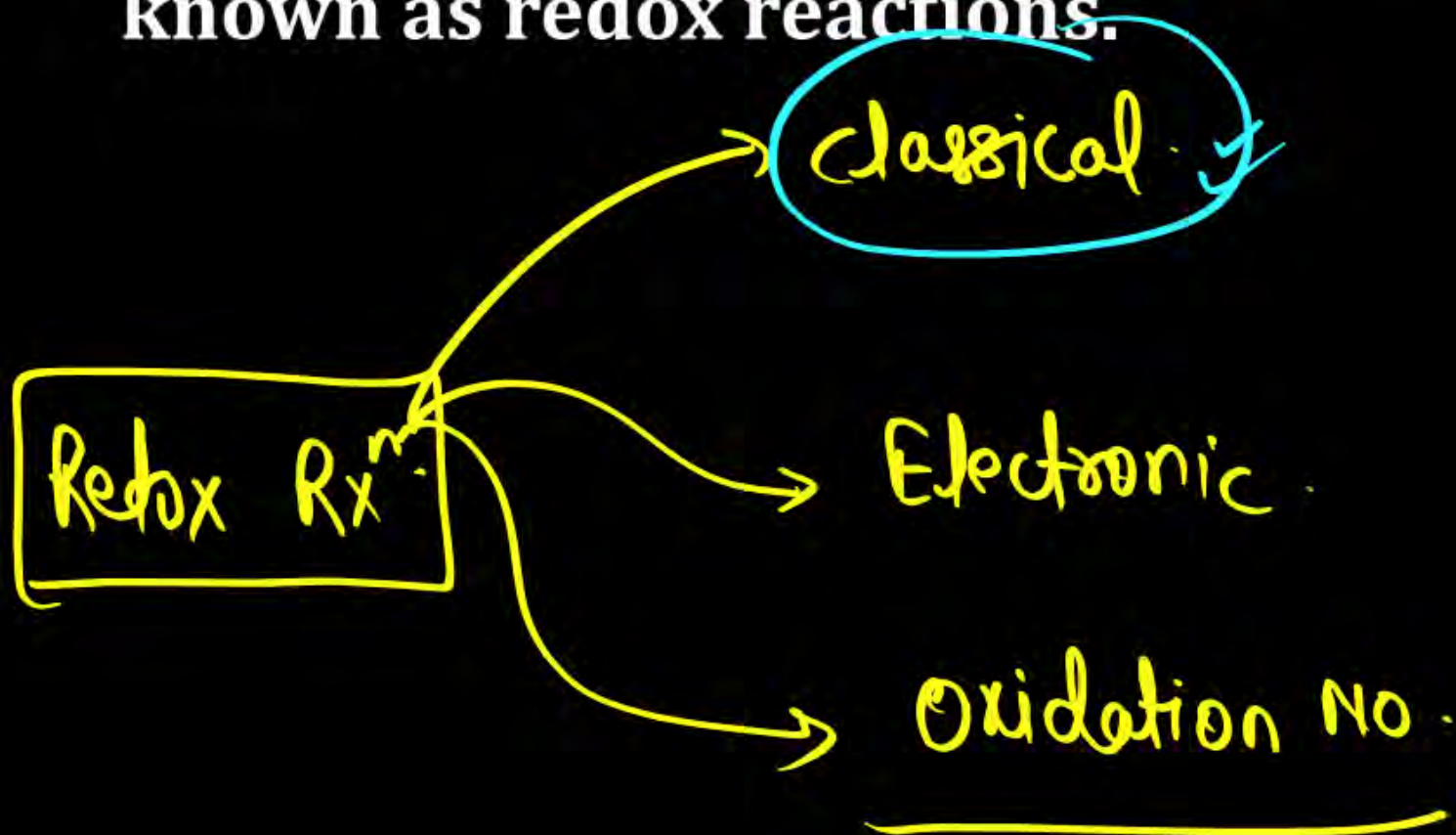




Introduction



- Reactions in which oxidation and reduction takes place simultaneously, are known as redox reactions.





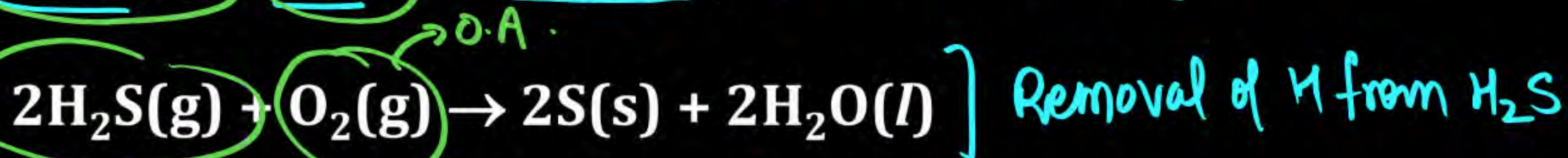
Classical Idea of Redox Reactions

Done



Oxidation is defined as the:

- addition of oxygen/electronegative element to a substance or
- removal of hydrogen/electropositive element from a substance.



Sub-oxidised

Handwritten note: Green arrow pointing from the underlined Mg in the third reaction to the word "Sub-oxidised".



Classical Idea of Redox Reactions

Done

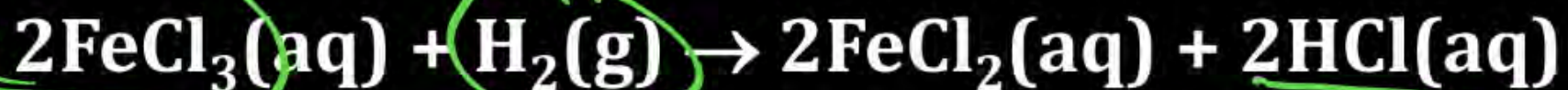


Reduction is defined as the

- Removal of oxygen/electronegative element from a substance or
- Addition of hydrogen or electropositive element to a substance.



(Removal of oxygen from mercuric oxide)



(removal of electronegative element, chlorine from ferric chloride)



(addition of hydrogen)

Sub. Reduced

R.A.



Classical Idea of Redox Reactions

Done



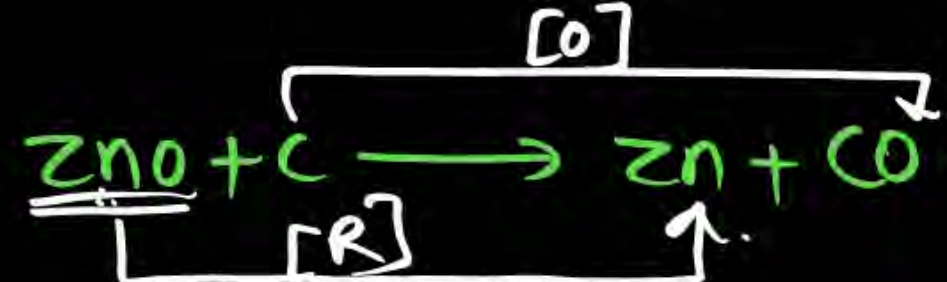
Reducing Agent or Reductant:

- A substance which supplies hydrogen or electropositive element
- Which removes oxygen or electronegative element.

Oxidising Agent or Oxidant:

- A substance which supplies oxygen or electronegative element
- Which removes hydrogen or electropositive element.

*



Sub. oxidised = C

Sub. Reduced = ZnO

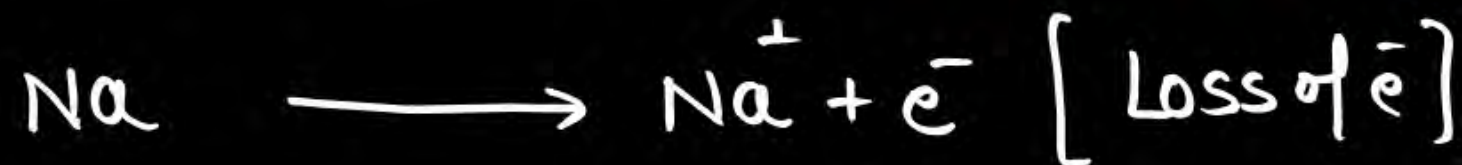
Oxidising agent = ZnO

Reducing agent = C



Electronic Concept of Redox Reactions

done



Sub. oxidised = Na } R.A.
Sub. reduced = Cl_2 } O.A.

Reducing agent $\rightarrow$ Loss of e^-
Oxidising agent $\rightarrow$ Gain of e^-

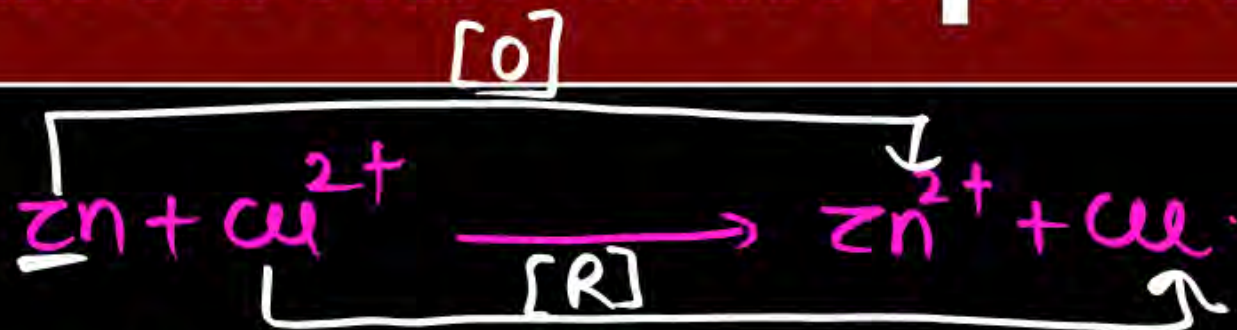
* Loss of e^- = Oxidation.

* Gain of e^- = Reduction.



Electronic Concept of Redox Reactions

Done

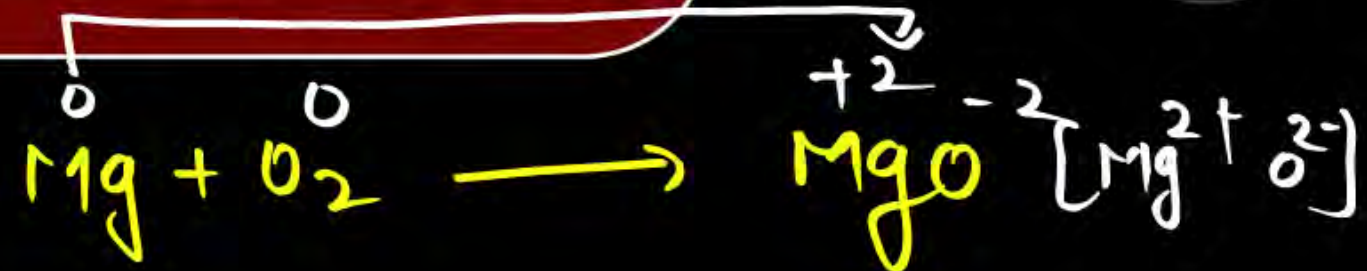


Sub. oxidised = Zn

Sub. reduced = Cu^{2+}

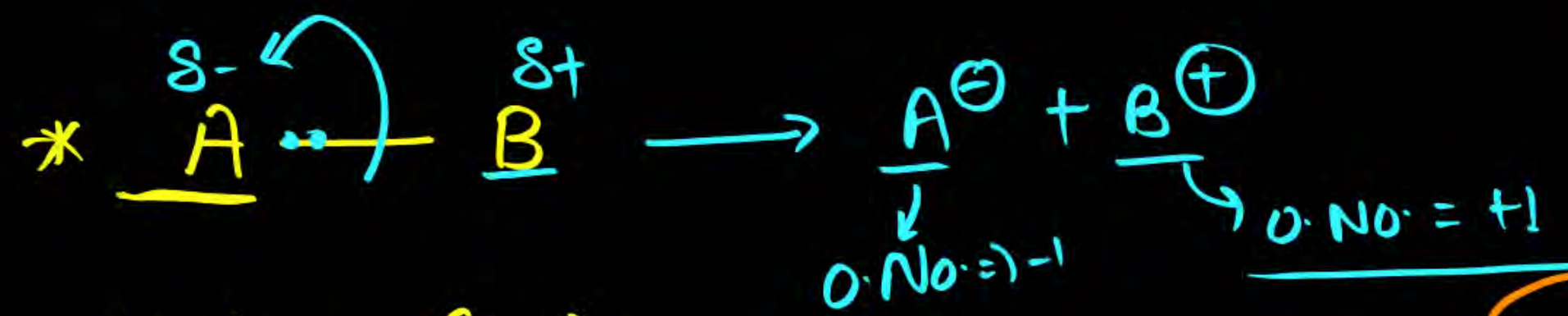
O.A. $\rightarrow \text{Cu}^{2+}$

R.A. $\rightarrow \text{Zn}$



Sub. oxidised = Mg $\rightarrow$ R.A.

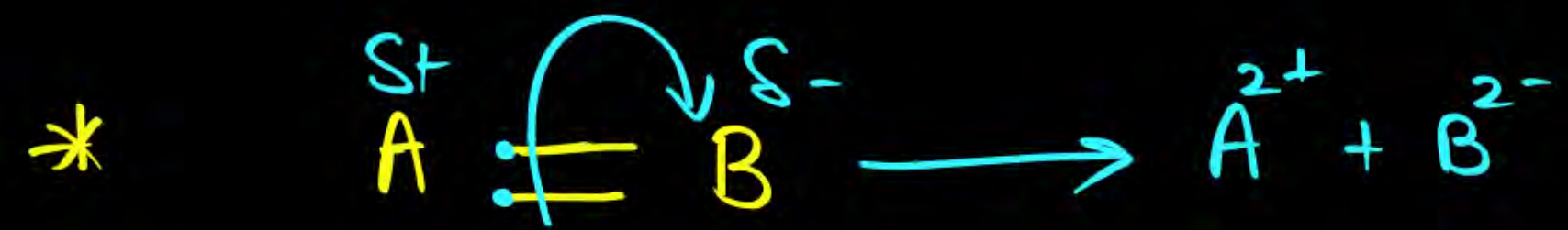
Sub. Reduced = $\text{O}_2 \rightarrow$ O.A.



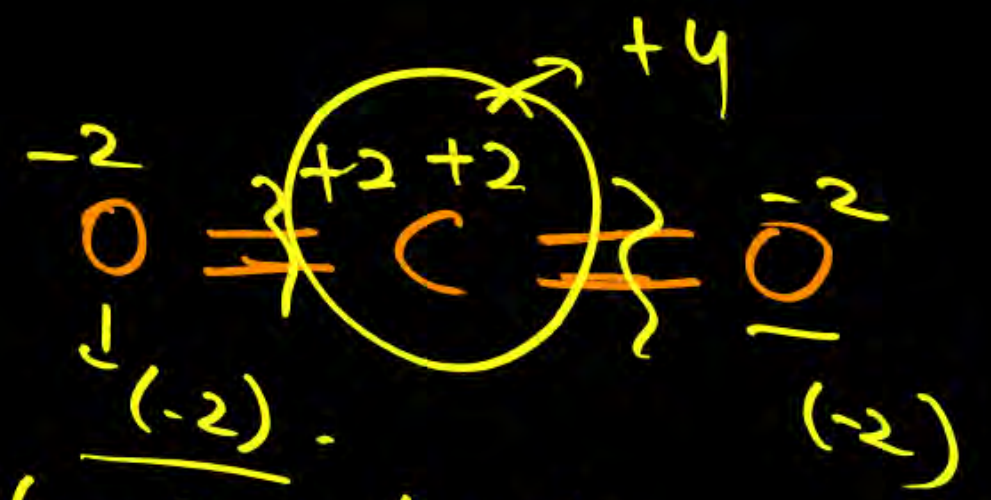
$(\text{E.N.})A > (\text{E.N.})B$



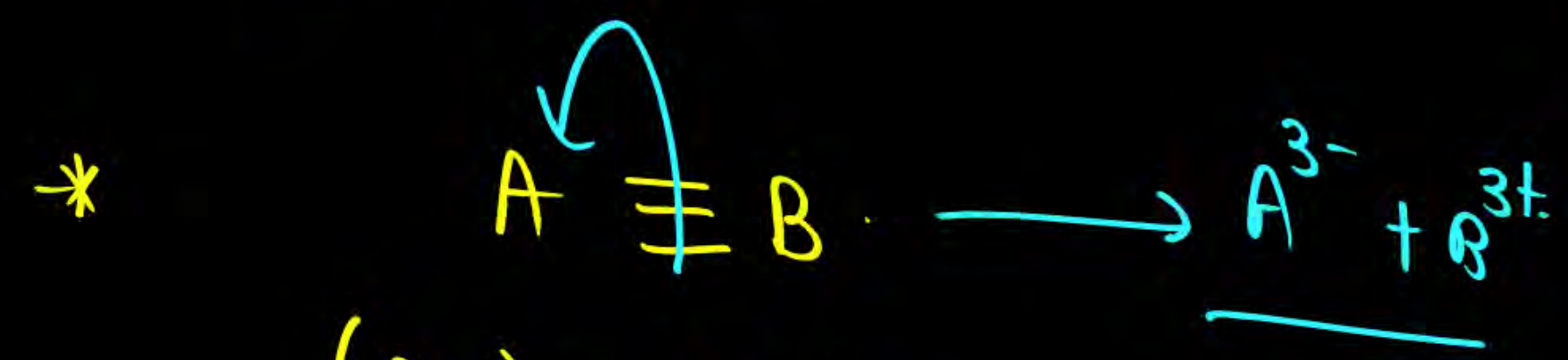
$(\text{E.N.})O > (\text{E.N.})H$



$(\text{E.N.})A < (\text{E.N.})B$



$(\text{E.N.})O > (\text{E.N.})C$



$(\text{E.N.})A > (\text{E.N.})B$



Oxidation Number

Def-1



- Oxidation number denotes the oxidation state of an element, is given by a set of rules on the basis that electron pair in a covalent bond belongs entirely to the most electronegative element.



Oxidation Number

Def-2



- It is defined as the charge which occurs on an atom when all the atoms bonded to it are removed from it in the form of their ions.



**Aaj ke liye
itna hi**

Physics Wallah



Books for JEE Mains



Books for JEE Mains +
Advanced (Combined)

The Ultimate JEE MAIN + ADVANCED COMBO



Chapterwise Theory
& Practice



Exam-Focused Revision
Support



NCERT-Based Solved
Questions



Detailed Explanations



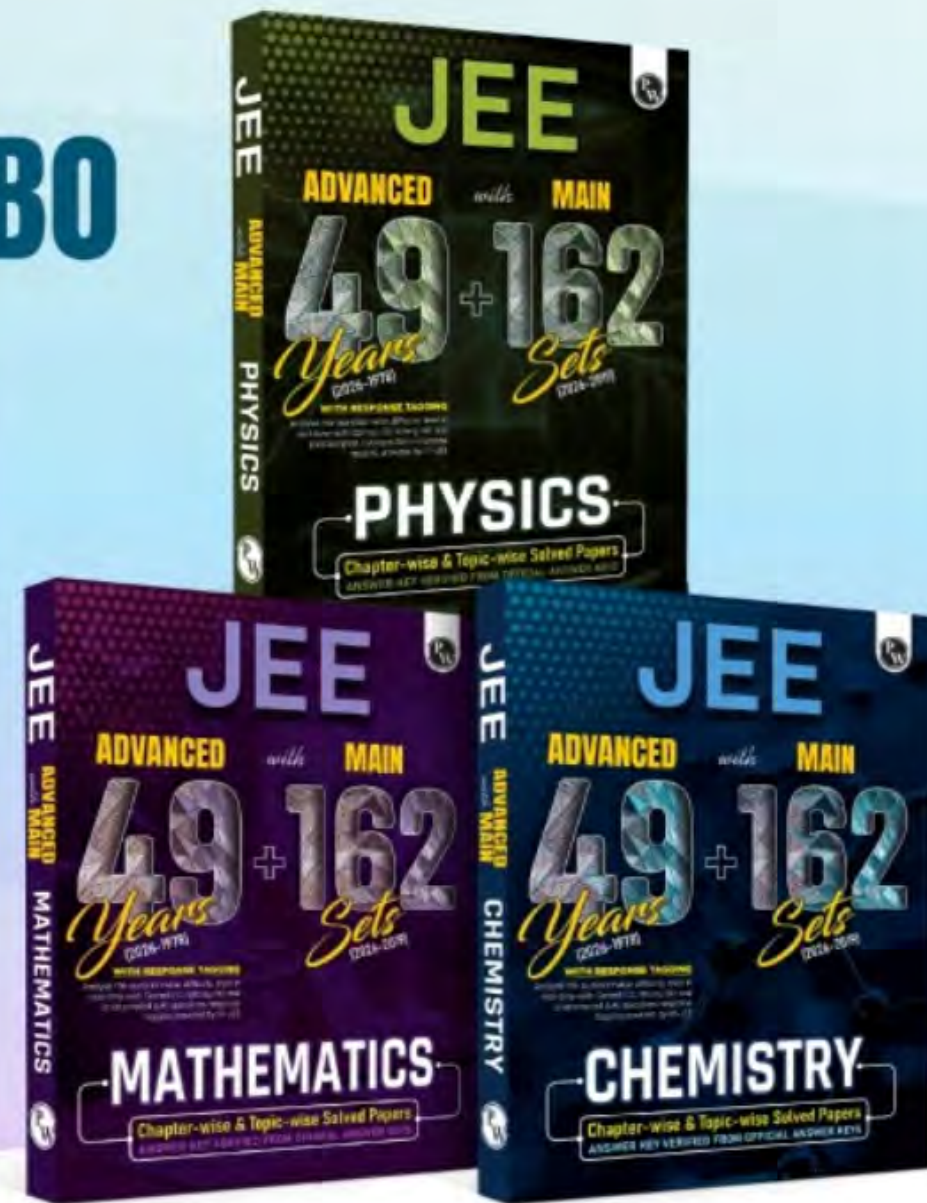
Competency-Based
Question Bank



Latest CBSE Pattern
Coverage



Buy Now!



Available on :-

amazon | Flipkart | PW Store



Homework



- Examples:
- Check your understanding:
- Topicwise :

* DPP → X.

* Revision.

* जो class में दिया है.

Thank

You